REZA BABAPIRI / ENGINEERING PORTFOLIO

DESIGN.
MODEL.
VALIDATE.

Mechanical design, embedded robotics, control engineering and vehicle simulation - documented technical work.

reza-babapiri.github.io

Mechanical Design & Manufacturing

Design and manufacturing evidence, including a documented Fusion 360 CAM workflow and professional industrial equipment experience. V6 and gearbox design will be added only after completion.

Industrial Expansion Joints | Design Practice
Bellows, flanges, piping components and mechanical design verification

Fusion 360 | CAD/CAM CNC Workflow
Aluminium 6061 milling plan with machining simulation and NC output

Embedded Systems & Robotics

Hardware-oriented work connecting ultrasonic sensing, Arduino control logic and motor actuation. Group contributions are described conservatively.

Autonomous Obstacle-Avoiding Robot
Arduino Uno, ultrasonic sensing, servo scanning and DC motor driving

Control Systems & Automation

Mathematical models, controller selection, simulation, limitations and results using MATLAB and Simulink.

Buck Converter | Cascaded PI Control
100 V to 48 V, 1 kW converter design with current and voltage feedback loops

Inverted Pendulum | LQR & LQI
Cart–pendulum stabilization, tracking, saturation and anti-windup

Vehicle Cruise Control | PI & Anti-Windup
Speed tracking under hills, actuator saturation and parameter changes

Kalman Filtering | Penalty-Kick Tracking
Tracking a moving target from noisy 2-D measurements

Automotive & Vehicle Simulation

Longitudinal and lateral dynamics, active steering, bicycle stability, and SUMO–Python–Simulink EV/HEV scenario studies.

EV / HEV Traffic Co-Simulation | Rome
SUMO traffic + Python bridge + MATLAB/Simulink vehicle models

Adaptive Cruise Control | PID vs Observer
Distance-error regulation and disturbance estimation

BMW 730 | Steering Dynamics
Speed-dependent stability, damping and steering response

Active Steering | Observer & Wiener Filter
PI feedback versus observer-based yaw-rate regulation

Bicycle Roll | Stability & Feedback
Speed-dependent roll dynamics and feedback stabilization

Computer Vision & AI

An early documented MATLAB image-processing activity. The previously described moss-detection project will be added as a complete case study when its code, images and evaluation outputs are provided.

MATLAB | Image Contrast Enhancement

Early computer vision / image-preprocessing exercise